

Cohort-Based CLV Analysis

Khristian Novoa
Turing College

Agenda

1. Business Context & Objective
2. Data & Definitions
3. Methodology
4. Cohort ARPU Heatmap
5. Cumulative Cohort Trends
6. Top vs. Weak Cohorts
7. Recommendations & Next Steps

Business Context & Objective

Context:

- Previous CLV models (e.g. Shopify's) exclude non-purchasing users → overstate customer value.
- Engagement drops off quickly; manager needs a 12-week view, not lifetime projections.

Objective:

- Compute true Average Revenue per User (ARPU) for weekly registration cohorts over 12 weeks, including all site visitors.
- Deliver actionable insights on which cohorts sustain the highest revenue and where to focus retention efforts.

Methodology

Define Registration Cohorts

- Extract each user's first event date (RegDate) and truncate to week → RegWeek.

Extract Purchases

- Filter on event_name = 'purchase' and purchase_revenue_in_usd > 0.
- Parse and truncate purchase dates to week → PurchaseWeek.

Build Cohort Data

- LEFT JOIN cohorts (FirstVisit) with Purchases on user_pseudo_id.
- Retain all users; non-purchasers default to \$0 revenue.

Aggregate Weekly Revenue

- GROUP BY RegWeek.
- COUNT cohort size (registrations).
- SUM Revenue for each week offset (Week0...Week12) via conditional aggregation.

Ensure 12-Week Window

- Include cohorts registered before 2021-01-31 to allow full Week0–Week12 tracking.

Cohort ARPU Heatmap

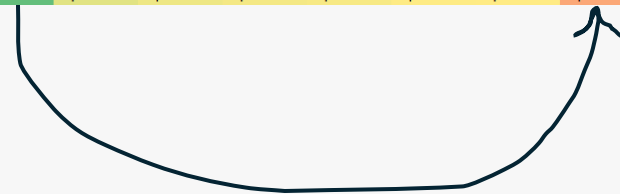
Average revenue per user by registration cohort and week (Week 0–12)

Key insights:

- **Peak in Week 0:** Cohorts average \$0.85 ARPU (early cohorts hit \$1.63).
- **Rapid drop-off:** ARPU falls $\geq 50\%$ by Week 1 and $> 80\%$ by Week 2.
- **Long-tail underperformance:** Weeks 6–12 ARPU plateaus below \$0.05.
- **Outlier:** 12/20/2020 cohort drops from \$1.00 $\rightarrow$ \$0.37 in Week 1.

Average Revenue Divided by Registrations Per Cohort

Week	Registrations	Week0	Week1	Week2	Week3	Week4	Week5	Week6	Week7	Week8	Week9	Week10	Week11	Week12
11/1/2020	20275	\$0.93	\$0.32	\$0.26	\$0.26	\$0.16	\$0.15	\$0.16	\$0.02	\$0.01	\$0.01	\$0.02	\$0.01	\$0.02
11/8/2020	16363	\$1.18	\$0.38	\$0.28	\$0.23	\$0.27	\$0.10	\$0.04	\$0.07	\$0.00	\$0.01	\$0.04	\$0.02	
11/15/2020	18062	\$1.37	\$0.29	\$0.22	\$0.22	\$0.17	\$0.03	\$0.03	\$0.02	\$0.02	\$0.01	\$0.00		
11/22/2020	19833	\$1.63	\$0.23	\$0.22	\$0.12	\$0.04	\$0.01	\$0.01	\$0.01	\$0.03	\$0.00			
11/29/2020	22099	\$1.31	\$0.36	\$0.24	\$0.05	\$0.01	\$0.02	\$0.01	\$0.01	\$0.01				
12/6/2020	28214	\$1.20	\$0.33	\$0.08	\$0.03	\$0.02	\$0.03	\$0.02	\$0.00					
12/13/2020	25239	\$1.00	\$0.11	\$0.04	\$0.03	\$0.04	\$0.03	\$0.00						
12/20/2020	17848	\$0.37	\$0.05	\$0.02	\$0.02	\$0.02	\$0.01							
12/27/2020	16550	\$0.34	\$0.05	\$0.00	\$0.02	\$0.01								
1/3/2021	22794	\$0.23	\$0.06	\$0.03	\$0.00									
1/10/2021	21479	\$0.40	\$0.06	\$0.01										
1/17/2021	20801	\$0.90	\$0.12											
1/24/2021	19561	\$0.19												
Average		\$0.85	\$0.20	\$0.13	\$0.10	\$0.08	\$0.05	\$0.04	\$0.02	\$0.01	\$0.01	\$0.02	\$0.02	\$0.02



Cumulative ARPU Growth

Cumulative average revenue per user by registration cohort (Weeks 0–12)

KEY INSIGHTS:

- **Strong initial lift:** Cohorts average 23.3% growth in Week 0.
- **Growth deceleration:** Slope drops from 23.3% → 8.4% by Week 3.
- **Long-tail plateau:** Weeks 7–12 grow only ~1.2%–1.6%, indicating low incremental value.
- **Top vs. bottom:** Early cohorts (e.g. 11/8/2020) maintain higher cumulative ARPU vs. post-12/20 cohorts.

Cumulative Average Per Cohort Week

Week	Registrations	Week0	Week1	Week2	Week3	Week4	Week5	Week6	Week7	Week8	Week9	Week10	Week11	Week12
11/1/2020	20275	\$0.93	\$1.25	\$1.52	\$1.78	\$1.93	\$2.09	\$2.25	\$2.27	\$2.28	\$2.30	\$2.32	\$2.33	\$2.35
11/8/2020	16363	\$1.18	\$1.56	\$1.84	\$2.07	\$2.34	\$2.45	\$2.48	\$2.55	\$2.55	\$2.56	\$2.60	\$2.62	
11/15/2020	18062	\$1.37	\$1.66	\$1.87	\$2.10	\$2.26	\$2.29	\$2.32	\$2.34	\$2.36	\$2.37	\$2.37		
11/22/2020	19833	\$1.63	\$1.86	\$2.09	\$2.21	\$2.24	\$2.26	\$2.26	\$2.27	\$2.31	\$2.31			
11/29/2020	22099	\$1.31	\$1.67	\$1.92	\$1.96	\$1.98	\$2.00	\$2.00	\$2.02	\$2.02				
12/6/2020	28214	\$1.20	\$1.52	\$1.61	\$1.64	\$1.66	\$1.69	\$1.71	\$1.71					
12/13/2020	25239	\$1.00	\$1.11	\$1.15	\$1.18	\$1.22	\$1.25	\$1.25						
12/20/2020	17848	\$0.37	\$0.42	\$0.44	\$0.47	\$0.48	\$0.49							
12/27/2020	16550	\$0.34	\$0.39	\$0.39	\$0.41	\$0.42								
1/3/2021	22794	\$0.23	\$0.29	\$0.32	\$0.32									
1/10/2021	21479	\$0.40	\$0.46	\$0.47										
1/17/2021	20801	\$0.90	\$1.02											
1/24/2021	19561	\$0.19												
Averages		\$0.85	\$1.05	\$1.18	\$1.28	\$1.36	\$1.40	\$1.44	\$1.47	\$1.48	\$1.49	\$1.51	\$1.53	\$1.54
Cumulative Growth			23.28%	12.24%	8.42%	6.38%	3.51%	2.73%	1.62%	0.93%	0.60%	1.40%	1.16%	1.18%

Top vs. Weak Cohorts

Week 12 ARPU: Top cohorts > \$2.00 vs. recent cohorts < \$0.60

KEY INSIGHTS:

- **Top cohorts (11/8–11/29/2020):** Week 12 ARPU $\geq$ \$2.00 per user
- **Middle cohorts (12/6/2020–1/3/2021):** Week 12 ARPU \$1.00–\$1.50
- **Recent cohorts (post-1/10/2021):** Week 12 ARPU \$0.35–\$0.60
- **Action:** Apply onboarding tactics from top cohorts to reengage underperforming segments

Week	Registrations	Week12
11/8/2020	16363	\$2.65
11/15/2020	18062	\$2.43
11/22/2020	19833	\$2.40
11/29/2020	22099	\$2.11
12/6/2020	28214	\$1.81
12/13/2020	25239	\$1.34
12/20/2020	17848	\$0.54
12/27/2020	16550	\$0.48
1/3/2021	22794	\$0.39
1/10/2021	21479	\$0.62
1/17/2021	20801	\$1.51
1/24/2021	19561	\$0.35

Recommendations & Next Steps

KEY RECOMMENDATIONS:

- **Revisit acquisition strategies:** Improve lead targeting to attract higher-quality users like the early cohorts.
- **Engage early:** Implement Week 0–3 onboarding and promotional tactics to boost initial ARPU when engagement is highest.
- **Focus on retention:** Develop re-engagement campaigns for Weeks 4–7 to extend the revenue tail and drive cumulative growth.
- **Monitor cohort performance:** Track ARPU by cohort weekly; iterate tactics for any cohort falling below \$1.00 ARPU by Week 4.